0.1 Previous Results

0.2 Event Selection

- Require either the Control Trigger (sometimes described as Minimum Bias Trigger), Multi Track (MT) or Electron Multi Track (eMT) which are described in Chapter **??**
- A vertex was selected if the total momentum was below $78/, GeV$, a total positive charge of 1, a χ^2 less than 40 and a vertex position between 105 and 180 m.
- The tracks associated to the vertex are selected if they are in acceptance of all four Straw spectrometer chambers and the CHOD. The track momentum must be greater than $5\,GeV$ and less than $60\,GeV$ with a $^+$ of less than 30. The tracks must traverse within 20 cm of each other.
- Require the presence of a three track in the vertex with a selected positive charged pion (π^+), and a selected electron positron pair ($\pi^\pm$). The pion is selected using missing mass between the Kaon momentum which is taken from the average beam momentum and the two positive tracks. If the positive track is within 5 sigma of the π^0 mass hypothesis then it is selected as a charged pion. If both positive tracks satisfy the condition the event is vetoed. From MC only approximately $0.1/, \%$ of Dalitz decays have both positive tracks satisfying the selection of the charged pion. The electron was assigned to the only negative track and the negative track which doesn't pass the charged pion selection was selected as the positron.

- A selected photon is also required. The missing momentum of the vertex was calculated with the Kaon momentum and the total vertex momentum and was used to extrapolate a straight line from the vertex position to the z plane of the LKr calorimeter. Within a radius of 15 cm of the extrapolated position a single cluster with a minimum energy of 5 GeV is selected as a photon from the decay. The photon energy has to satisfy the condition of being within 5 GeV of the calculated missing vertex momentum.

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The analysis started using the same procedure used in the NA48/2 [**?**]